True or false? (unless otherwise specified.)

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1. **Degree of precision.** The degree of precision of the following quadrature rule is 3:

$$\int_0^1 f(x)\,\mathrm{d}x \approx \frac{f(0)+f(1)}{2} - \frac{f'(1)-f'(0)}{12}\,.$$

2. **Degree of precision.** The degree of precision of Simpson's 3/8 rule,

$$\int_0^3 f(x)\,\mathrm{d}x \approx \frac{3}{8}\Big(f(0)+3f(1)+3f(2)+f(3)\Big),$$

is equal to 4.

3. **Degree of precision.** If the quadrature rule

$$\int_{-1}^1 f(x)\,\mathrm{d}x \approx \sum_{i=1}^n w_i f(x_i)$$

has degree of precision at least 0, then $w_1 + \cdots + w_n = 2$.

4. **Composite quadrature.** Suppose that $f \in C^\infty[a,b]$ and let $I_n[f]$ denote the approximation of the integral $I[f] \coloneqq \int_a^b f(x)\,\mathrm{d}x$ obtained with the composite midpoint rule with n cells, that is

$$I_n[f] = h\sum_{i=0}^{n-1} f\left(x_i + \frac{h}{2}\right), \qquad h = \frac{b-a}{n}, \qquad x_i = a + ih\,.$$

Then there exists a constant $C \in (0,\infty)$ such that

$$\forall n \in \{1,2,\dots\}, \qquad \Big|I[f] - I_n[f]\Big| \leqslant \frac{C}{n^2}\,.$$

Hint: Compute the degree of precision of the midpoint rule $\int_{-1}^1 f(x)\,\mathrm{d}x \approx 2f(0)$, and use the theorem relating the degree of precision of a quadrature rule to the order of convergence, with respect to h, of the associated composite rule.

5. **Composite quadrature.** With the notation of the previous item, it holds that $I_n[f] = I[f]$ for all $n \geqslant 1$ whenever f is a polynomial of degree at most 2.

6. **Composite quadrature.** Let $f \in C^\infty[a,b]$. For large n, doubling the number of cells in the composite Simpson rule reduces the error by approximately a factor 8.

7. **Composite quadrature.** Let $f\colon [a,b] \to \mathbb{R}$ be a convex function, and let T_n denote the composite trapezoidal approximation of $\int_a^b f(x)\,\mathrm{d}x$ with n cells. Then $T_n \geqslant \int_a^b f(x)\,\mathrm{d}x$ for all $n \geqslant 1$.

8. **Convergence estimate.** Let $f\colon [0,2] \to \mathbb{R}$ be the discontinuous function

$$f(x) = \begin{cases} 0 & \text{if } x < \sqrt{2}, \\ 1 & \text{if } x \geqslant \sqrt{2}, \end{cases}$$

and let T_n denote the composite trapezoidal approximation of $\int_0^2 f(x)\,\mathrm{d}x$ with n cells. Then there exists C such that $|\int_0^2 f(x)\,\mathrm{d}x - T_n| \leqslant C/n^2$ for all $n \geqslant 1$.

9. **Gaussian integration.** The two-point Gauss–Legendre rule on $[-1,1]$ has nodes $\pm 1/\sqrt{3}$ and both weights equal to 1.

10. **Gaussian integration.** All the weights of the n-point Gauss–Legendre rule are positive.

11. **Gaussian integration.** The n-point Gauss–Legendre rule is exact for all polynomials of degree up to $2n$.

12. **Richardson extrapolation.** Let $T(h)$ be the composite trapezoidal approximation of $\int_a^b f(x)\,\mathrm{d}x$ with step size h. Then the approximation
$$\frac{4T(h) - T(2h)}{3}$$
coincides with the composite Simpson approximation with step size h.

13. **Gaussian integration.** There exist nodes x_1, x_2 and weights w_1, w_2 such that the rule
$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx w_1 u(x_1) + w_2 u(x_2)$$
has a degree of precision equal to 3.

14. **Richardson extrapolation.** Let $f \in C^\infty[a,b]$ and let $S(h)$ denote the composite Simpson approximation of $\int_a^b f(x)\,\mathrm{d}x$ with step size h. Assuming that $S(h) = I + c_4 h^4 + c_6 h^6 + O(h^8)$, it holds that
$$\frac{16S(h) - S(2h)}{15} = I + O(h^6)\,.$$

15. **Julia.** In the following code, `R(f, a, b, n)` returns the composite Simpson approximation of $\int_a^b f(x)\,\mathrm{d}x$ with $2n$ cells:

```
function T(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    return h * (sum(f, x[2:end-1]) + (f(a) + f(b)) / 2)
end

R(f, a, b, n) = 2T(f, a, b, 2n) - T(f, a, b, n)
```

16. **Bonus.** Let $f \in C^\infty(\mathbb{R})$ and $x \in \mathbb{R}$ be such that $f''(x) \neq 0$, and let $D(h)$ denote the forward finite difference approximation of $f'(x)$:
$$D(h) = \frac{f(x+h) - f(x)}{h}\,.$$
Then the approximation
$$\frac{D(h) + D(2h)}{2}$$
satisfies $\left|\frac{D(h)+D(2h)}{2} - f'(x)\right| = O(h^2)$ as $h \to 0$. *Justify briefly.*